

Policy Iteration for Multiplicative Noise Output Feedback Control

Benjamin Gravell, Matilde Gargiani,
John Lygeros, and Tyler Summers



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So you thought linear systems were easy?

Consider two complicating factors

Random dynamics

$$x_{t+1} = A_t x_t + B_t u_t + w_t$$

Random partial observability

$$y_t = C_t x_t + v_t$$

A_t, B_t, C_t, w_t, v_t are iid random

Limited view of internal state

Multiplicative Noise Output Feedback Optimal Control (MOFOC)

$$\begin{aligned} & \underset{\pi}{\text{minimize}} && \lim_{T \rightarrow \infty} \mathbb{E} \frac{1}{T} \sum_{t=0}^{T-1} \begin{bmatrix} x_t \\ u_t \end{bmatrix}^\top \begin{bmatrix} Q_{xx} & Q_{xu} \\ Q_{ux} & Q_{uu} \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix} \\ & \text{subject to} && \begin{bmatrix} x_{t+1} \\ y_t \end{bmatrix} = \begin{bmatrix} A_t & B_t \\ C_t & 0 \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix} + \begin{bmatrix} w_t \\ v_t \end{bmatrix} \end{aligned}$$

Complicating factors **interact**:

Optimal linear policy is a **coupled** controller and state estimator

separation principle

Prior work:

- Value iteration for MOFOC [1] - *slow*
- Policy iteration for LQG [2, 3, 4] - *no multiplicative noise*

This work:

Policy iteration for MOFOC

Who cares?

- 1 Solve MOFOC problems faster
- 2 Open the door to novel approximate policy iteration algorithms

Linear Dynamic Controller (F, K, L)

$$\hat{x}_{t+1} = F\hat{x}_t + Ly_t \quad (\text{state estimator})$$

$$u_t = K\hat{x}_t \quad (\text{feedback control})$$

Using (F, K, L) , the closed-loop augmented state dynamics are

$$\begin{bmatrix} x_{t+1} \\ \hat{x}_{t+1} \end{bmatrix} = \underbrace{\begin{bmatrix} A_t & B_t K \\ LC_t & F \end{bmatrix}}_{\Phi'_t} \begin{bmatrix} x_t \\ \hat{x}_t \end{bmatrix} + \begin{bmatrix} I_n & 0 \\ 0 & L \end{bmatrix} \begin{bmatrix} w_t \\ v_t \end{bmatrix}$$

Why restrict search to linear policies?

Optimal policy in this class can be computed from Riccati equations.

Definition 1 (Mean-square stability [1])

The second moment of the augmented state converges to a finite positive semidefinite matrix, i.e.

$$\lim_{t \rightarrow \infty} \mathbb{E} \begin{bmatrix} x_t \\ \hat{x}_t \end{bmatrix} \begin{bmatrix} x_t \\ \hat{x}_t \end{bmatrix}^T = S' \succeq 0$$

Definition 2 (Mean-square compensatability [1])

There exists a controller which renders the system ms-stable.

Assumption: The system is mean-square compensatable.

If (F, K, L) is ms-stabilizing, the **generalized discrete-time Lyapunov equations (GDLYAP)** have well-defined solutions:

$$P' = \mathbb{E} [\Phi_t'^T P' \Phi_t'] + Q' \quad (\text{steady-state value function})$$

$$S' = \mathbb{E} [\Phi_t' S' \Phi_t'^T] + W' \quad (\text{steady-state second moment})$$

How to solve GDLYAP?

- Vec-kron trick to convert to a system of linear equations
- Specialized Krylov subspace techniques [5]

Later we need an array of four matrices

$$X = (P, \hat{P}, S, \hat{S})$$

which picks and adds certain blocks of (P', S')

Gain operators

$$\mathcal{K}(X) := -\mathcal{G}_{uu}^{-1}(X)\mathcal{G}_{ux}(X),$$

$$\mathcal{L}(X) := \mathcal{H}_{xy}(X)\mathcal{H}_{yy}^{-1}(X).$$

generate an improved policy based on the value array X

$\mathcal{G}(\cdot)$ and $\mathcal{H}(\cdot)$ are affine operators

- Represent the “Q-function” that characterizes the cost of taking a control action + state estimate update while being in a given augmented state, then following (F, K, L) forever after
- Too big to fit on this slide

Solve the coupled nonlinear matrix Riccati equations [1]

$$\mathcal{R}(X) := \begin{pmatrix} -P + \mathcal{G}_{xx}(X) - \mathcal{G}_{xu}(X)\mathcal{G}_{uu}^{-1}(X)\mathcal{G}_{ux}(X) \\ -\hat{P} + \mathcal{E}(X) + \mathcal{G}_{xu}(X)\mathcal{G}_{uu}^{-1}(X)\mathcal{G}_{ux}(X) \\ -S + \mathcal{H}_{xx}(X) - \mathcal{H}_{xy}(X)\mathcal{H}_{yy}^{-1}(X)\mathcal{H}_{yx}(X) \\ -\hat{S} + \mathcal{F}(X) + \mathcal{H}_{xy}(X)\mathcal{H}_{yy}^{-1}(X)\mathcal{H}_{yx}(X) \end{pmatrix} = 0$$

where

$$\mathcal{E}(X) := (A - \mathcal{L}(X)C)^\top \hat{P} (A - \mathcal{L}(X)C)$$

$$\mathcal{F}(X) := (A + BK(X))\hat{S}(A + BK(X))^\top$$

Then the optimal policy (F^*, K^*, L^*) is

$$K^* = \mathcal{K}(X)$$

$$L^* = \mathcal{L}(X)$$

$$F^* = A + BK^* - L^*C$$

Value iteration [1] - state-of-the-art since 1992, but *slow*

$$X^{k+1} = X^k + \mathcal{R}(X^k)$$

Motivating question

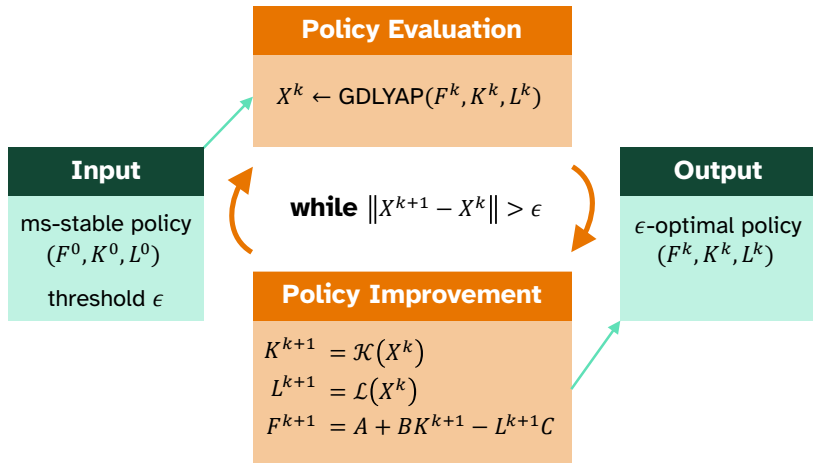
Can we solve the optimal policy design equations more quickly?

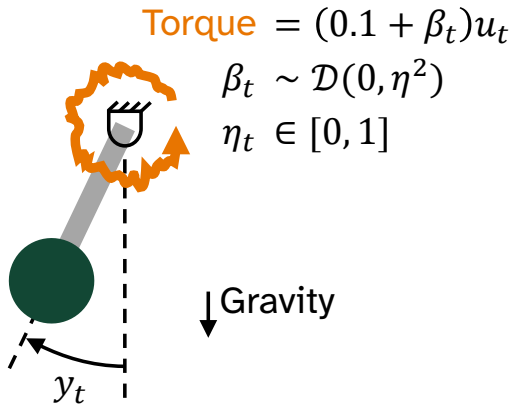
Key ideas:

Policy iteration is an instance of the provably rapidly convergent Newton's method in the fully observed setting [6, 7]

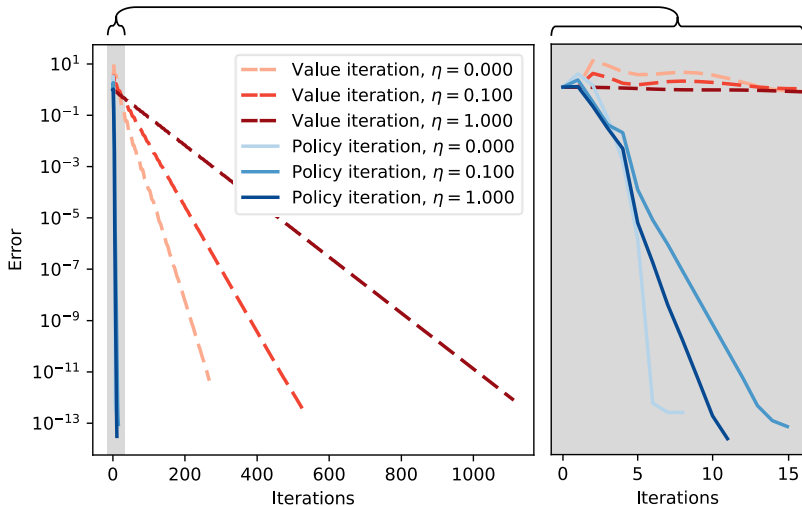
Policy evaluation and policy improvement steps can be generalized to the partially observed setting

Stands to reason that generalizing policy iteration to the partially observed setting would yield rapid convergence

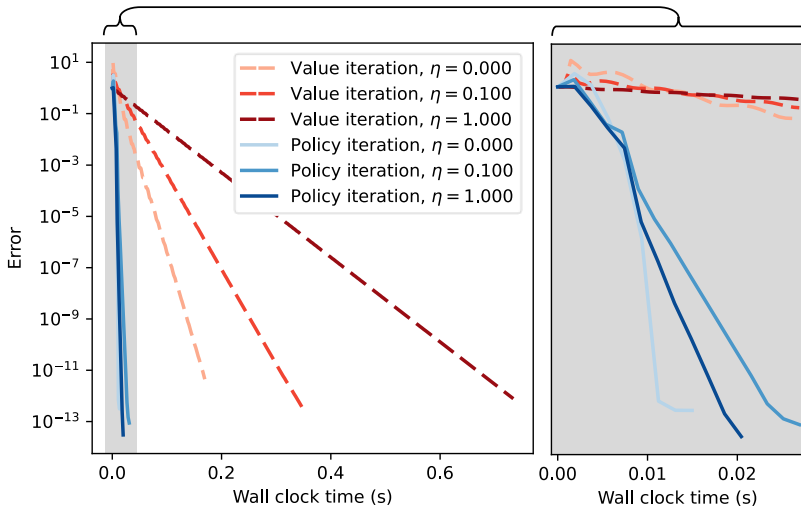




Experiment: pendulum system, iterations



Experiment: pendulum system, clock time



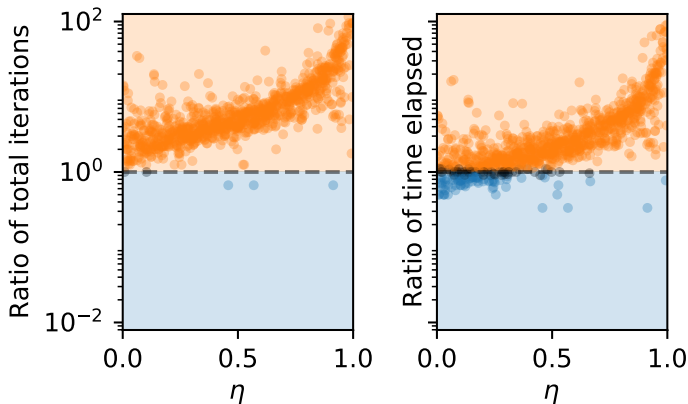
2 states, 1 input, 1 output

Problem data randomly generated from standard normal distributions

System matrices and noise levels scaled to get open-loop ms-stability
→ Initial policy $(A, 0, 0)$ is valid

Noise level η drawn randomly between 0 and 1 (critically ms-stable)

Value iteration vs policy iteration



Summary

We proposed a policy iteration algorithm for multiplicative noise optimal output feedback control design that, empirically, converges far faster than value iteration.

Future work

- Formalize connection to Newton method & leverage this view to prove a quadratic rate of convergence like in [8, 9, 10]
- Investigate novel approximate policy iteration algorithms that use input-output data to approximately execute policy evaluation and policy improvement steps to solve POMDPs.

Python code available at

<https://github.com/TSummersLab/policy-iteration-mlqc>

Contact information:

Ben Gravell

bjgravell@gmail.com

Matilde Gargiani

gmatilde@ethz.ch

John Lygeros

jlygeros@ethz.ch

Tyler Summers (PI)

tyler.summers@utdallas.edu



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